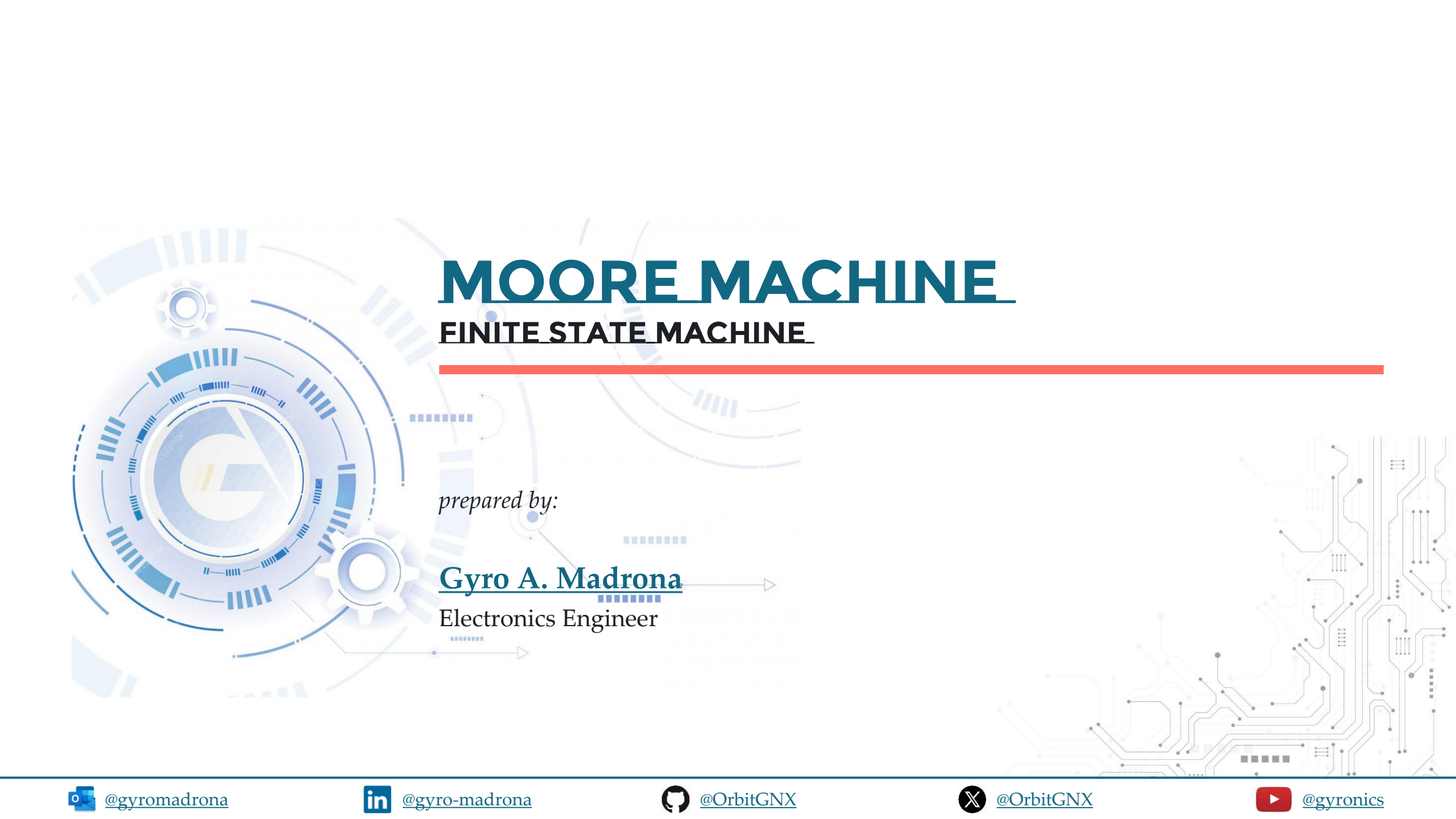




MOORE MACHINE

FINITE STATE MACHINE

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TOPIC OUTLINE

Moore Machine

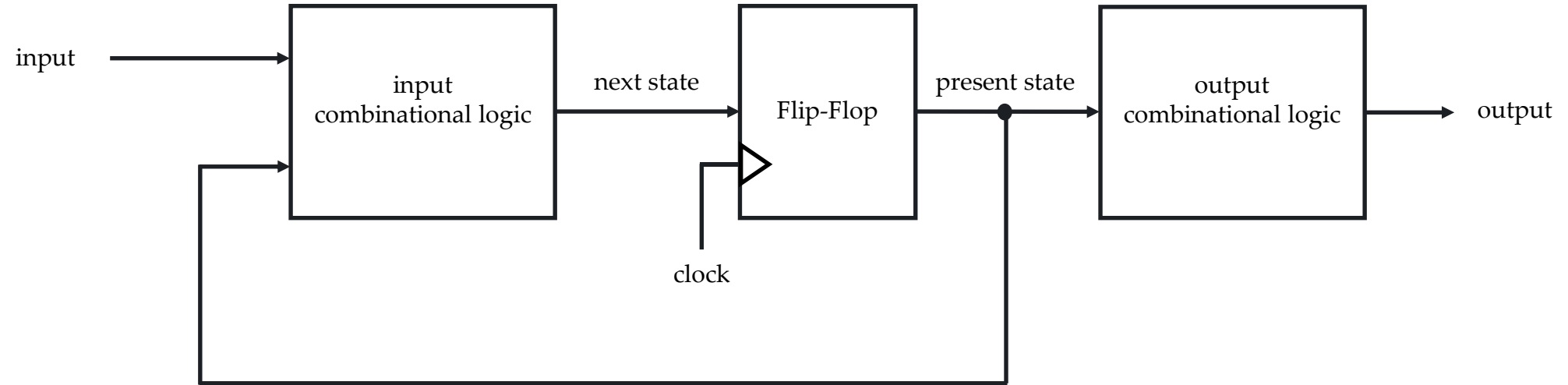
- Sequence 101
- Sequence 110
- Sequence 1011



MOORE MACHINE



MOORE MACHINE



A Moore machine is a type of finite-state machine in which the outputs depend only on the present state.



SEQUENCE 101

This Moore FSM detects the non-overlapping sequence 101, assuming the input stream arrives LSB-first.

System Level Design

```
module seq_101( );
```

```
endmodule
```



SEQUENCE 101

State Diagram



SEQUENCE 110

This Moore FSM detects the non-overlapping sequence 110, assuming the input stream arrives LSB-first.

System Level Design

```
module seq_110( );
```

```
endmodule
```



SEQUENCE 110

State Diagram



SEQUENCE 1011

This Moore FSM detects the non-overlapping sequence 1011, assuming the input stream arrives LSB-first.

System Level Design

```
module seq_1011( );
```

```
endmodule
```



SEQUENCE 1011

State Diagram



SEQUENCE 1011

This Moore FSM detects the overlapping sequence 1011, assuming the input stream arrives LSB-first.

System Level Design

```
module seq_1011( );
```

```
endmodule
```



SEQUENCE 1011

State Diagram



LABORATORY

